

PAVAN KUMAR BUDATI

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Professional Summary

Lead Full Stack Developer with hands-on experience in **Java (8/21)**, **Spring Boot**, **Microservices**, and **React JS**, delivering scalable, secure, and high-performance enterprise applications across telecom payments, healthcare, banking, logistics, and trading execution domains. Deep expertise in **OOP principles**, **Spring framework internals**, **encryption/decryption security protocols**, **RESTful API design**, and cloud deployments using **AWS**, **Helm**, and **GitLab CI/CD**. Proven track record of independently delivering quality work products, optimizing transaction pipelines, and accelerating engineering productivity using modern **GenAI tools** (Claude, GitHub Copilot).

Technical Skills

System Design & Architecture: High-Level Design (HLD), Low-Level Design (LLD), Microservices Architecture, Event-Driven Architecture, Scalability & High Throughput, Distributed Systems, Caching Strategies, Fault Tolerance

Languages: Java (8/21) — Streams, Lambdas, Collections, Threads, Virtual Threads; JavaScript, TypeScript, Python, HTML, CSS

Frameworks: Spring Boot, Spring MVC, Spring Security, Spring Cloud, Spring Webflux, Hibernate, JPA, JDBC, Servlets, Node.js/Express, FastAPI

Frontend: React JS (TypeScript, Hooks, State Management, Component Design), React Native (Expo), Tailwind CSS, Vite, ES6+, Moment.js

Security & Encryption: AES-256, RSA Encryption/Decryption, Spring Security, JWT, OAuth 2.0, OpenID Connect (OIDC), RBAC

Microservices & Integration: RESTful APIs, SOAP Services, Apache Kafka, WebSockets, Inter-Process Communication (IPC), Async Messaging, API Gateway

Databases: PostgreSQL, MS SQL Server, MongoDB, Redis, SQLite, Transaction Management, Query Optimization

Cloud & DevOps: AWS (EC2, S3, Lambda, RDS, DynamoDB, API Gateway), Helm, Kubernetes, GitLab CI/CD, GitHub Actions, Docker, Jenkins

Engineering Practices: OOP, SOLID Principles, Design Patterns, TDD/JUnit, Agile/Scrum, High-Precision Scheduling

AI Tools: Claude, GitHub Copilot, GitHub Copilot Spaces

Experience

UST Global

May 2026 – Present

Lead I - Software Engineer — Telecom & Payments Domain

Hyderabad, India

- **Architecting and developing** resilient backend microservices and responsive **React JS** frontends for high-throughput telecom payment processing and transaction settlement systems.
- **Implemented robust security protocols** using enterprise-grade **encryption and decryption techniques** (AES/RSA) to protect sensitive billing data and cardholder information in compliance with PCI-DSS standards.
- **Engineered scalable REST APIs** using **Java 21** and **Spring Boot**, streamlining payment gateway integrations and subscription billing workflows.
- **Containerized and deployed** cloud-native microservices on **AWS** utilizing **Helm** charts for Kubernetes orchestration, simplifying multi-environment release automation.
- **Established automated CI/CD pipelines** using **GitLab**, enforcing rigorous code scanning, automated testing, and zero-downtime deployment strategies.
- **Leveraged Claude AI** to assist in rapid technical design prototyping, refactoring legacy payment logic, and generating comprehensive unit/integration test suites.

Vitech Systems Asia

Oct 2025 – Apr 2026

Java Full Stack Developer — Consultant

Hyderabad, India

- **Designed** scalable backend microservices and **RESTful APIs** for Vitech's core enterprise product **Velocity**, applying advanced **system design**, **OOP principles**, and **Spring Boot** with **PostgreSQL**.
- **Leveraged GitHub Copilot** and **Copilot Spaces** to auto-generate **JUnit test cases**, maintaining **80%+ unit test coverage** across backend services.
- **Built** dynamic, reusable **React JS** components using **Hooks** and custom logic modules to accelerate UI development.
- **Deployed** and managed microservices on **AWS EC2**, configuring **security groups**, **IAM roles**, and **Auto Scaling** policies.
- **Utilized AWS Lambda**, **API Gateway**, **S3**, and **RDS (PostgreSQL)** for event-driven serverless workflows and secure document storage.

Endava Solutions (Formerly GalaxE Solutions)

Jul 2022 – Oct 2025

Senior Java Full Stack Developer — GxCapture — Healthcare & Banking

Bangalore, India

- **Led** end-to-end full-stack development on **GxCapture**, delivering modules using **Java 21**, **Spring Boot**, **Spring Security**, **GraphQL**, and **React JS**.
- **Implemented Reactive Programming** (Spring Webflux, R2DBC) to handle concurrent data loads, achieving a **50% reduction in page load time**.

- **Integrated Virtual Threads (Java 21)**, delivering a **2x throughput improvement** across critical backend workflows.
- **Designed JWT-based** authentication and authorization flows with **Spring Security** and **RBAC** for healthcare and banking APIs.

Java Full Stack Developer — GxQuality — Low-Code Test Platform

- **Built** a containerized mainframe test automation framework (**Java**, **Docker**, **Socket Programming**, **X3270 emulator**), enabling execution preview across **100% of test cases**.
- **Contributed** to a low-code testing platform supporting Web, REST, ETL, and Mainframe using **SOAP/REST APIs** and **Kafka**.

Exner Technologies

Mar 2021 – Jun 2022

Developer Trainee

Hyderabad, India

- **Contributed** to enterprise product development using **Java**, **Spring Framework**, **SQL**, and **JDBC**.

Key Projects

Neo-Copier — Multi-Account Algorithmic Trade Copier & Execution Engine [Live Demo](#) | [Frontend](#) | [Backend](#)

- **Architected a full-stack trade replication engine** enabling sub-second order distribution from Master accounts to multiple Child accounts across **Kotak Neo**, **Upstox**, and **MT5** broker APIs.
- **Engineered asynchronous multi-account dispatching** using **Java 21 Virtual Threads** / **CompletableFuture** and **Python FastAPI AsyncIO**, supporting dynamic lot scaling (fixed lot, ratio-based, and capital-weighted sizing).
- **Designed an automated OCO (One-Cancels-the-Other) bracket engine** with real-time order status tracking via **WebSockets**, automatically auto-canceling counter-leg orders to eliminate position risk.
- **Implemented high-speed Scrip Master indexing**, parsing 100,000+ derivative option contracts (NFO/BSE) into in-memory lookup maps for microsecond symbol-to-token resolution.
- **Developed cross-platform frontends** using **React 19 (TypeScript, Vite, Tailwind CSS, Lightweight Charts)** for real-time web monitoring and **React Native (Expo)** for mobile position tracking and panic square-off execution.

Low Latency Forex Order Scheduler

[Live Demo](#) | [Frontend](#) | [Backend](#)

- **Engineered** a high-precision algorithmic order scheduler using **React (TypeScript)**, **Node.js**, **Python (FastAPI)**, and **MetaTrader 5 (MT5)** for sub-millisecond execution accuracy during high-volatility news events.
- **Architected a 2-stage scheduling engine** combining coarse sleep timers with a **high-resolution CPU Spin-Lock** (`process.hrtime.bigint()`), eliminating Node.js event-loop jitter (5–15ms) to achieve precise execution.
- **Designed direct Inter-Process Communication (IPC)** with active MT5 desktop instances, enabling local zero-credential order execution without external cloud broker API dependencies.
- **Implemented real-time broker clock synchronization** to calculate server time offsets and empirical execution drift (0ms -- 2ms), alongside role-based execution isolation (**Admin Mode** vs. **Guest Sandbox**).

Telecom Payments Encryption & Gateway Integration — UST

- **Designed** and integrated end-to-end payload encryption/decryption mechanisms for telecom billing services; automated deployment pipelines with **GitLab CI/CD** and **Helm** on AWS Kubernetes clusters.

Reactive Microservices for High-Volume Data Optimization — GxCapture

- **Designed** reactive **Spring Boot** microservices using **Webflux** and **R2DBC**; implemented **async inter-service communication** via **Kafka**; achieved **50% page load reduction** and significantly improved backend scalability.

Spring Security + JWT API Platform with RBAC — Multi-Client

- **Built** reusable **Spring Security** authentication/authorization framework with **JWT**, **RBAC**, and custom claims across healthcare (**GxCapture**), logistics (**USPS**), and automotive (**Volkswagen**) client platforms.

MainframeDriver — Selenium-Architecture Automation Library for Mainframe — GxQuality

- **Reverse-engineered** the **Selenium WebDriver** architecture and built **MainframeDriver** — a full Java automation library that mirrors Selenium's design patterns (**Driver interface**, **Command Executor**, **RemoteWebDriver model**) targeting IBM Mainframe terminals via **x3270 emulator** over **Socket/Telnet protocol**.
- **Integrated Java 21 Virtual Threads** for lightweight, high-concurrency session handling, enabling hundreds of parallel sessions with minimal memory overhead and **100% test case traceability**.

Education

Jawaharlal Nehru Technical University

July 2018 – June 2022

B.Tech — Electronics and Communication Engineering (ECE)

CGPA: 7.76